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ELEC 3200: System Modeling, Analysis and Control

Lecture 10: Simulation and Implementation of Systems

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Control Systems and MATLAB

- **Polynomial representation** One can represent a polynomial $p(s) = p_0 s^n + p_1 s^{n-1} + \cdots + p_n$ by a vector $\boldsymbol{p} = \begin{bmatrix} p_0 & p_1 & \cdots & p_n \end{bmatrix}$.

$$p(s) = s^3 + 2s^2 + 3s + 4, \quad q(s) = 5s + 6.$$

In MATLAB, we have

```
>> p=[1 2 3 4];
>> q=[5 6];
```

⇒ cannot add together! !!

  - The **roots** of $p(s)$

```
>> roots(p)
```

  - The **sums** of $p(s)$ and $q(s)$

```
>> p+[0 0 q]
```

  - The **products** of $p(s)$ and $q(s)$

```
>> conv(p,q)
```

https://ww2.mathworks.cn/help/matlab/polynomials.html

Control Systems and MATLAB

- **State space representation** Suppose a system described by its state space model $\boldsymbol{F}$:

$$\dot{\boldsymbol{x}}(t) = \begin{bmatrix} 0 & 1 \\ -1 & -2 \end{bmatrix} \boldsymbol{x}(t) + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u(t)$$

$$y(t) = \begin{bmatrix} 2 & 1 \end{bmatrix} \boldsymbol{x}(t)$$

In MATLAB, we have

```
>> A=[0 1; -1 -2];
>> B=[0 ; 1];
>> C=[2 1];
>> D=0;
>> F=ss(A,B,C,D)
```

https://ww2.mathworks.cn/help/control/ref/ss.html

Control Systems and MATLAB

- **Transfer function representation** Suppose a system described by its transfer function model $\boldsymbol{G}$:

$$\frac{s+2}{s^2+2s+1}$$

In MATLAB, we have

```
>> num=[1 2];
>> den=[1 2 1];
>> G=tf(num,den)
```

- **Convert from one to another**

```
>> F=tf(F);
>> G=ss(G);
```

transfer back to state space model

```
>> F=ss(F); %different from the original state space
```

But correct!

Control Systems and MATLAB

- **Give back the parameters of a system**

```
>> [A,B,C,D]=ssdata(F);
>> [num,den]=tfdata(F);
```

- **Poles and zeros of a system**

```
>> zero(F);
>> pole(F);
```

- **The sum and product of systems**

```
>> F+G;
>> F*G;
>> F-G;
>> F/G; %except G(s) = 0
```

Control Systems and MATLAB

- **Another way of representing polynomials** Consider a polynomial $p(s) = p_0 s^n + p_1 s^{n-1} + \cdots + p_n$ by a vector $\boldsymbol{p} = \begin{bmatrix} p_0 & p_1 & \cdots & p_n \end{bmatrix}$.

$$p(s) = s^3 + 2s^2 + 3s + 4, \quad q(s) = 5s + 6.$$

Simply by interpreting them as transfer functions with 1 as the denominator polynomials

```
>> p=tf([1 2 3 4],1);
>> q=tf([5 6],1);
```

  - The **sums** of $p(s)$ and $q(s)$

```
>> p+q
```

  - The **products** of $p(s)$ and $q(s)$

```
>> p*q
```

Control Systems and MATLAB

- **Negative feedback connection of two systems**

$$\frac{F}{1+FG}$$

In MATLAB, we have

```
>> feedback(F,G)
```

- **Positive feedback**

$$\frac{F}{1-FG}$$

In MATLAB, we have

```
>> feedback(F,G,1)
```

https://ww2.mathworks.cn/help/control/ref/inputoutputmodel.feedback.html

Control Systems and MATLAB

- **Impulse response**

```
>> [y,t]=impulse(G);
```

- **Step response**

```
>> [y,t]=step(G);
```

- **General input signal response** Use a MATLAB command `lsim`. For example, the sinusoidal response of the system:

```
>> t=0:0.1:10;
>> u=sin(t);
>> y=lsim(G,u,t);
```

Hardware Simulation and Implementation

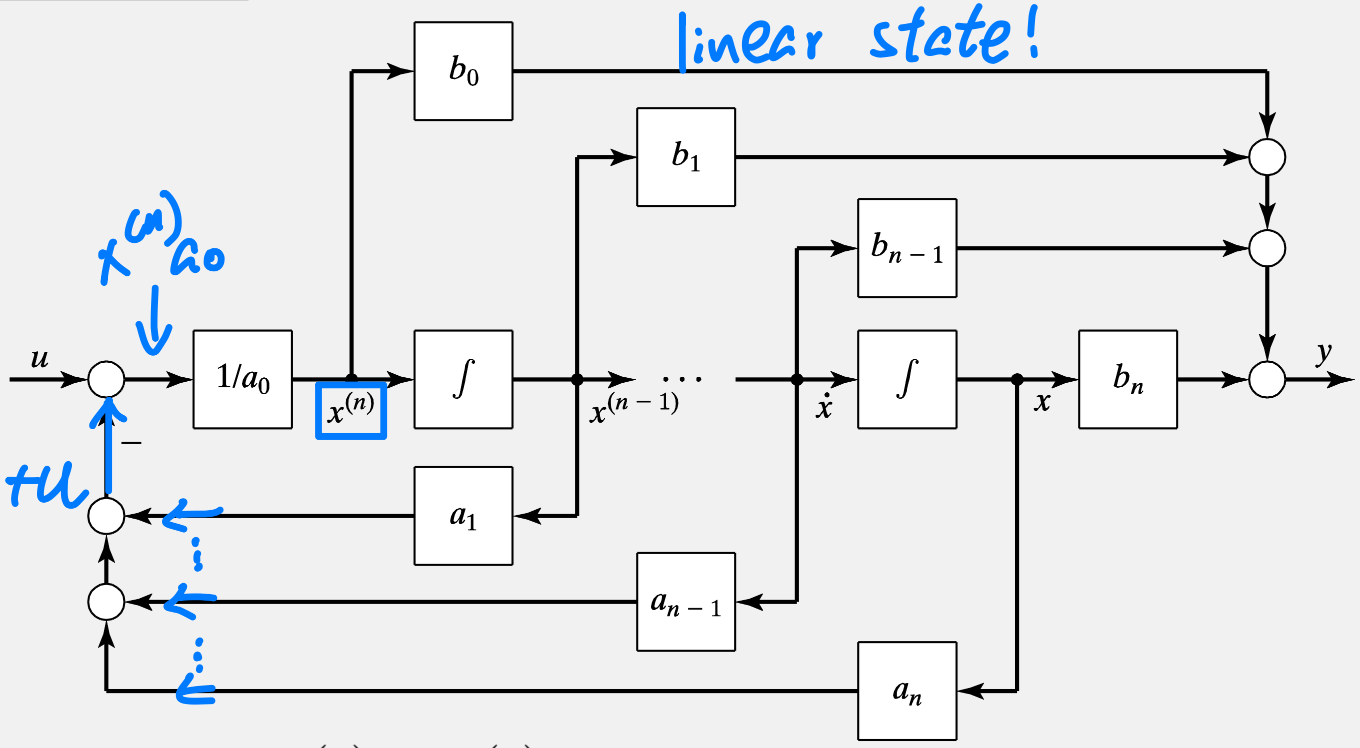
- **Modeling** Physical systems $\Longrightarrow$ Transfer function/State space
- **Realization** Physical systems $\Longleftarrow$ Transfer function/State space

- **Hardware simulation and implementation**
  - Let a system be given by a proper nth order transfer function $G(s)$.
  - Find the **realization** of the system by the **op-amp circuit**.

$$G(s) = \frac{b(s)}{a(s)} = \frac{b_0 s^n + b_1 s^{n-1} + \cdots + b_n}{a_0 s^n + a_1 s^{n-1} + \cdots + a_n}, \quad a_0 \neq 0.$$

- **Two types of realization**
  - Controller form
  - Observer form

Controller Form Realization



$a_0 x^{(n)} = u - (\quad \cdots \quad)$
$= u - (a_1 x^{(n-1)} + a_2 x^{(n-2)} + \cdots + a_{n-1}\dot{x} + a_n x)$

$a_0 s^n X = -a_1 s^{n-1} X - a_2 s^{n-2} X - \cdots - a_n X + u$

- **Controller form realization**
  - Notice that

$$a_0 x^{(n)}(t) = -a_1 x^{(n-1)}(t) - \cdots - a_n x(t) + u(t).$$

Take the Laplace transforms:

$$a_0 s^n X(s) = -a_1 s^{n-1} X(s) - \cdots - a_n X(s) + U(s) \Longrightarrow a(s)X(s) = U(s).$$

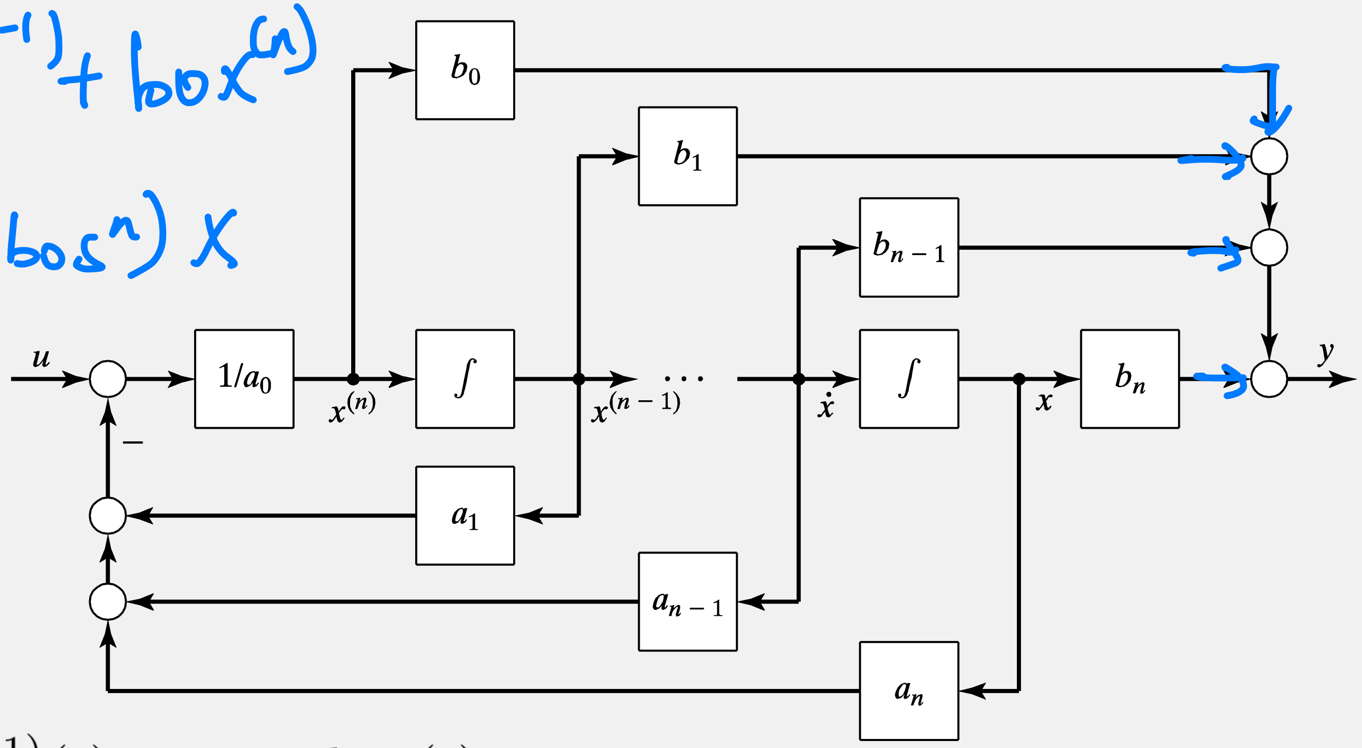
$[a_0 s^n + a_1 s^{n-1} + \cdots + a_n] X = u$ $\qquad \frac{X}{u} = \frac{1}{\cdots}$

Controller Form Realization

$y = b_n x + b_{n-1} x^{(1)} + \cdots + b_1 x^{(n-1)} + b_0 x^{(n)}$

$y = (b_n + b_{n-1} s + \cdots + b_1 s^{n-1} + b_0 s^n) X$

$\frac{y}{x} = \cdots$



- **Controller form realization**
  - Also, notice that

$$y(t) = b_0 x^{(n)}(t) + b_1 x^{(n-1)}(t) + \cdots + b_n x(t).$$

Take the Laplace transforms:

$$Y(s) = b_0 s^n X(s) + b_1 s^{n-1} X(s) + \cdots + b_n X(s) = b(s)X(s) \Longrightarrow \frac{Y(s)}{U(s)} = \frac{b(s)}{a(s)}.$$

Controller Form Realization

- **State space model of controller form**

Assign the voltages across capacitors as state variables, then $\boldsymbol{x}(t) = \begin{bmatrix} x^{(n-1)}(t) \\ \vdots \\ \dot{x}(t) \\ x(t) \end{bmatrix}$.

The corresponding state space model is

$$\boxed{\begin{aligned} \dot{\boldsymbol{x}}(t) &= \begin{bmatrix} -\frac{a_1}{a_0} & \cdots & -\frac{a_{n-1}}{a_0} & -\frac{a_n}{a_0} \\ 1 & \cdots & 0 & 0 \\ \vdots & \ddots & \vdots & \vdots \\ 0 & \cdots & 1 & 0 \end{bmatrix} \boldsymbol{x}(t) + \begin{bmatrix} \frac{1}{a_0} \\ 0 \\ \vdots \\ 0 \end{bmatrix} u(t) \\ y(t) &= \begin{bmatrix} b_1 - b_0\frac{a_1}{a_0} & \cdots & b_{n-1} - b_0\frac{a_{n-1}}{a_0} & b_n - b_0\frac{a_n}{a_0} \end{bmatrix} \boldsymbol{x}(t) + \frac{b_0}{a_0} u(t). \end{aligned}}$$

Observer Form Realization

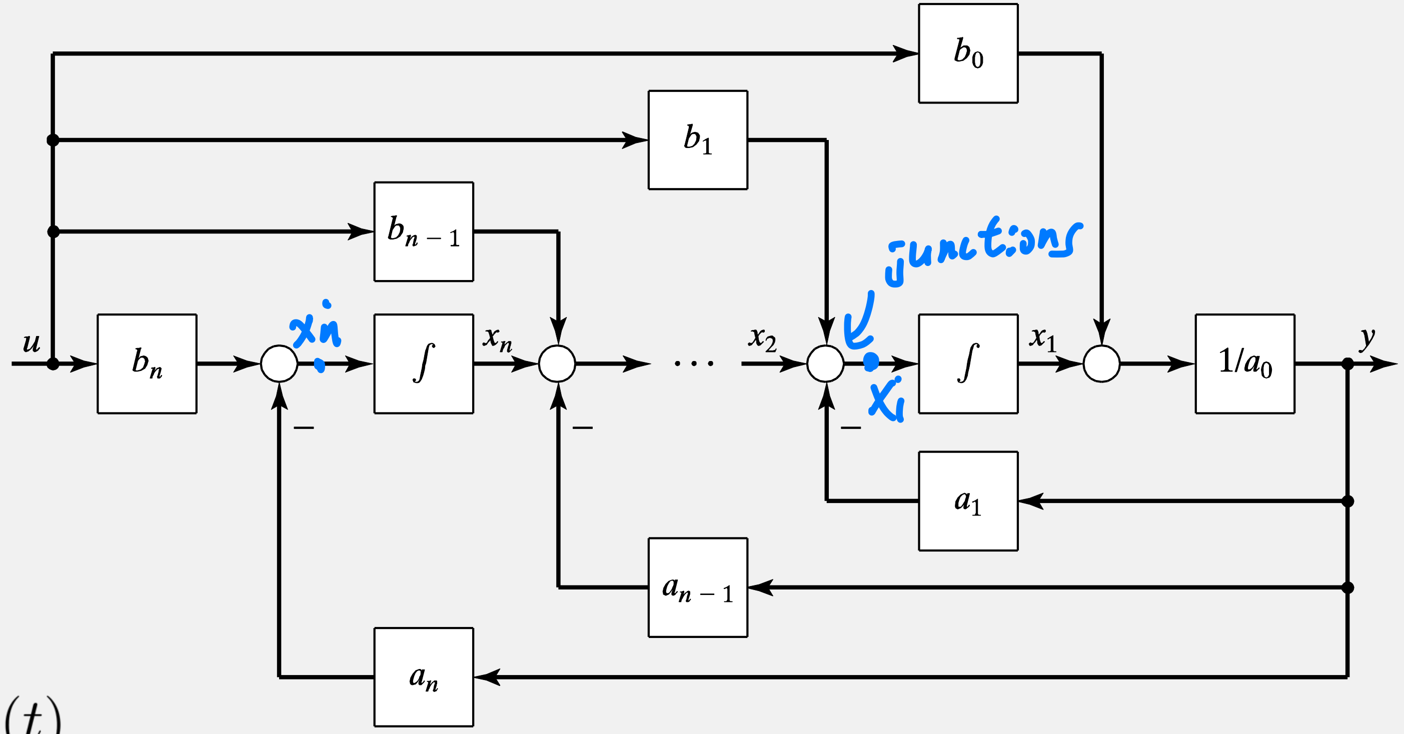
$a_0 y = b_0 u + x_1$

$\dot{x}_1 = x_2 - a_1 y + b_1 u$

$\vdots$

$\dot{x}_n = -a_n y + b_n u$

Amplifiers



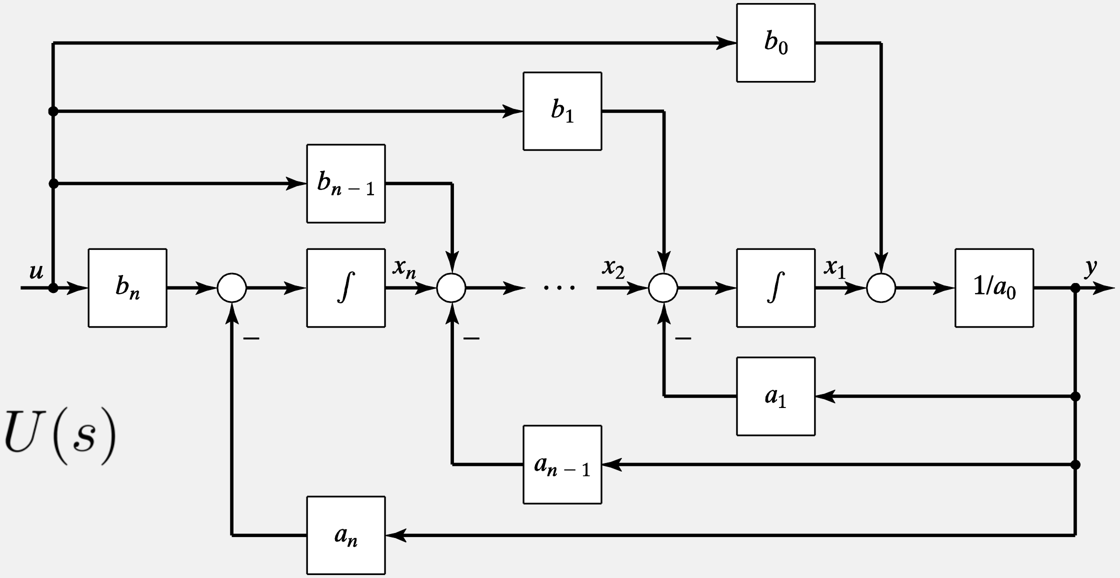
- **Observer form realization**
  - Notice that

$$
\begin{aligned}
a_0 y(t) &= x_1(t) + b_0 u(t) \\
\dot{x}_1(t) &= x_2(t) - a_1 y(t) + b_1 u(t) \\
&\vdots \\
\dot{x}_{n-1}(t) &= x_n(t) - a_{n-1} y(t) + b_{n-1} u(t) \\
\dot{x}_n(t) &= -a_n y(t) + b_n u(t).
\end{aligned}
$$

Observer Form Realization

$a_0 s^n Y = -a_1 s^{n-1} Y + b_0 s^n U + b_1 s^{n-1} U$

- **Observer form realization**



  – Take the Laplace transforms:

Multiply

① — s^n

② — s^{n-1}

$\vdots$

s

1

$$
\begin{aligned}
a_0 s^n Y(s) &= s^n \cancel{X_1(s)} + s^n b_0 U(s) \\
+\ s^n \cancel{s X_1(s)} &= s^{n-1} \cancel{X_2(s)} - s^{n-1} a_1 Y(s) + s^{n-1} b_1 U(s) \\
+ &\ \ \vdots \\
s^2 \cancel{s X_{n-1}(s)} &= s \cancel{X_n(s)} - s\, a_{n-1} Y(s) + s\, b_{n-1} U(s) \\
+\ \cancel{s X_n(s)} &= -a_n Y(s) + b_n U(s).
\end{aligned}
$$

  – Multiply the above equations by $s^n, s^{n-1}, \ldots, s, 1$, respectively, and add them altogether:

$$a(s)Y(s) = b(s)U(s) \Longrightarrow \frac{Y(s)}{U(s)} = \frac{b(s)}{a(s)}.$$

Observer Form Realization

- **State space model of observer form**

  Assign the voltage across capacitors as state variables, then

  $\boldsymbol{x}(t) = \begin{bmatrix} x_1(t) & x_2(t) & \cdots & x_n(t) \end{bmatrix}^T.$

  The corresponding state space model is

$$\dot{\boldsymbol{x}}(t) = \begin{bmatrix} -\dfrac{a_1}{a_0} & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ -\dfrac{a_{n-1}}{a_0} & 0 & \cdots & 1 \\ -\dfrac{a_n}{a_0} & 0 & \cdots & 0 \end{bmatrix} \boldsymbol{x}(t) + \begin{bmatrix} b_1 - b_0 \dfrac{a_1}{a_0} \\ \vdots \\ b_{n-1} - b_0 \dfrac{a_{n-1}}{a_0} \\ b_n - b_0 \dfrac{a_n}{a_0} \end{bmatrix} u(t)$$

$$y(t) = \begin{bmatrix} \dfrac{1}{a_0} & 0 & \cdots & 0 \end{bmatrix} \boldsymbol{x}(t) + \frac{b_0}{a_0} u(t).$$

Review & Exercise

- How to use Matlab for simulation?
- What’s the controller form?
- What’s the observer form?

Appendix

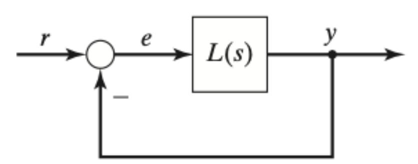
Simulink

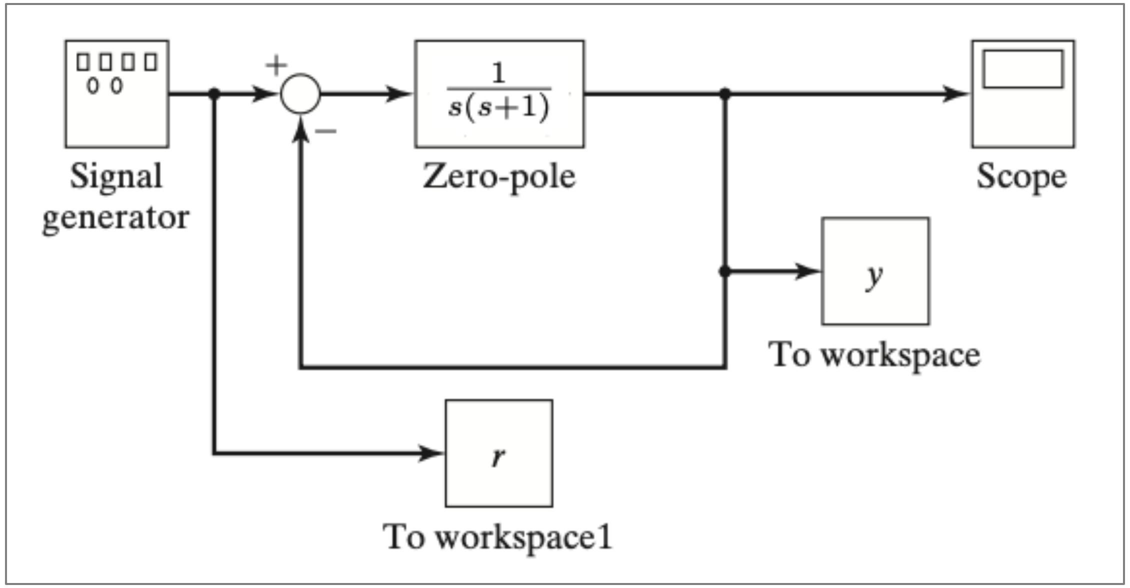
https://ww2.mathworks.cn/en/solutions/control-systems.html?s_tid=srchtitle_site_search_2_control%20diagram

Another software product associated with MATLAB is SIMULINK. It can be used to simulate an interconnected system, represented by a block diagram.

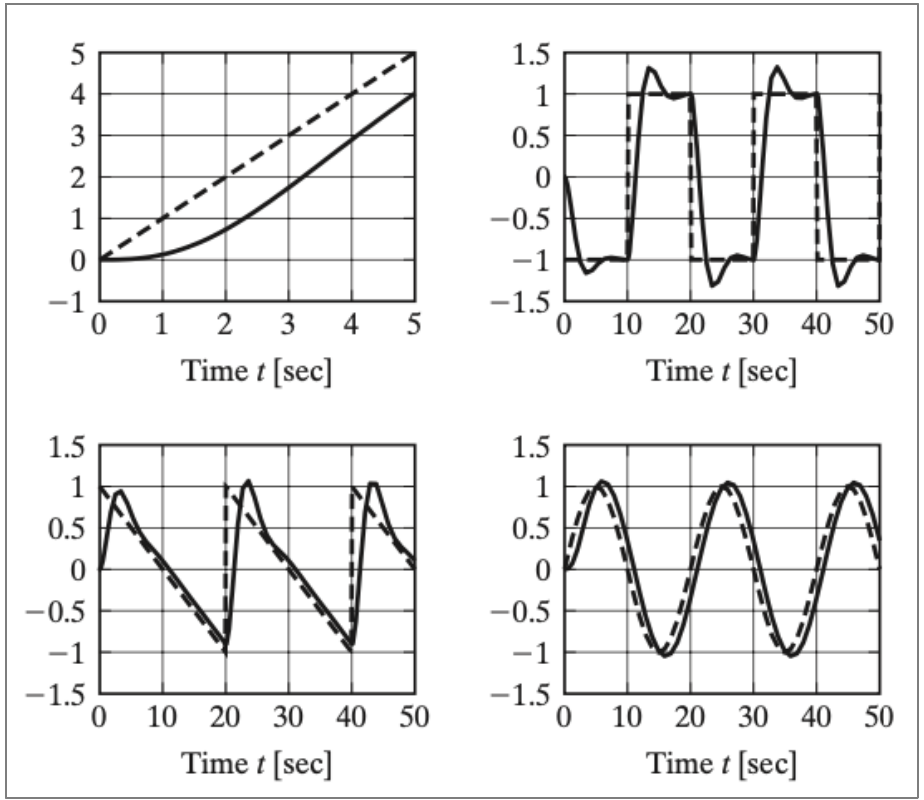
Example:

$$L(s) = \frac{1}{s(s+1)}.$$



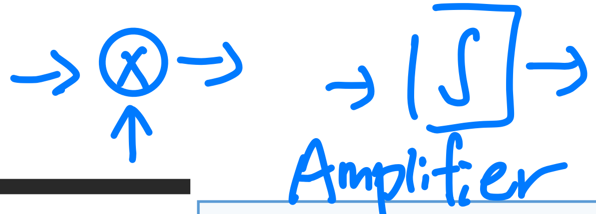


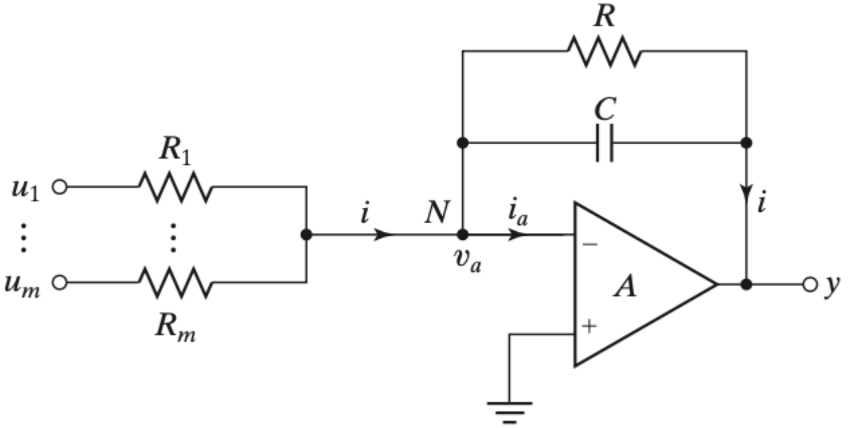
SIMULINK diagram



Responses to a ramp signal, a square wave, a sawtooth signal, and a sine wave

op-amp circuit





Owing to the infinite input impedance of the op-amp, the current flowing into the op-amp is zero, i.e., $i_a(t) = 0$. Thus we have

$$i(t) = \frac{u_1(t)}{R_1} + \frac{u_2(t)}{R_2} + \cdots + \frac{u_m(t)}{R_m}$$

$$i(t) = -\frac{1}{R}y(t) - C\frac{dy(t)}{dt}.$$

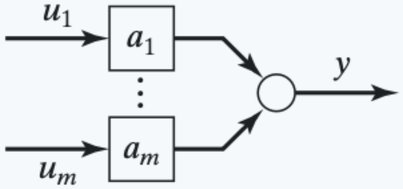
Hence

$$RC\frac{dy(t)}{dt} + y(t) = -\left[\frac{R}{R_1}u_1(t) + \cdots + \frac{R}{R_m}u_m(t)\right].$$

If $C = 0$, i.e., the capacitor is disconnected, then

$$y(t) = -\left[\frac{R}{R_1}u_1(t) + \cdots + \frac{R}{R_m}u_m(t)\right],$$

i.e., the output is a negatively weighted sum of the inputs. In this case, the circuit is called an (inverting) summing amplifier. If further, $m = 1$, the circuit is called an (inverting) amplifier.



If $R = \infty$, i.e., the resistor is disconnected, then

$$y(t) = -\left[\frac{1}{R_1C}\int_0^t u_1(\tau)d\tau + \cdots + \frac{1}{R_mC}\int_0^t u_m(\tau)d\tau\right],$$

i.e., the output is a negatively weighted sum of the integrals of the inputs. If further, $m = 1$, then the output is proportional to the integral of the input. In this case, the circuit is called an (inverting) integrator.